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# SDC Hackathon

# Technical Tools Declaration

**Hosted by Xiamen University Malaysia (XMUM) AI Club**

**Submission Deadline: 25 October 2026**

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

***Instagram:*** @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

**SDC HACKATHON 2026**

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## 1. Team & Project Overview

| | |
|---|---|
| **Team Name:** | Really cool team name |
| **Project / Solution Name:** | Unbundle |
| **Name (Leader):** | Amina Zaatir |
| **Email & Phone (Leader):** | za.amina2005@gmail.com +216 29 459 029 |
| **University Affiliation(s):** | Xiamen University Malaysia |
| **Live Prototype URL:** | https://unbundle.replit.app/ |
| **Code Repository URL:** | https://github.com/tsetsen-oyun/SDC-Hackaton-2026 |

## 2. Architecture & Technology Stack

### 2.1 System Architecture Overview

*Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.*

***[Maximum 200 words]***

**Offline data build:** public sources (O*NET, Anthropic Economic Index, METR, ESCO, UN SDG API, SDSN, DOSM, US BLS, Microsoft Learn and Coursera catalogues) are cleaned by the team into dated CSV snapshots kept in Git. A language model (Qwen3-32B-AWQ, Runpod public endpoint) drafted job-to-SDG links once, offline; they are reviewed by the team.

**Runtime:** the React 19 and Vite web app calls a REST API (Node 22, Express 5). The API loads the newest snapshot into memory and passes it to a plain JavaScript engine (forecast, match, route, mentor, skills). There is no database server and nothing is stored on the server. Two outside calls exist: (1) optional free-text intake, where the typed text goes to the Runpod model, which may only pick tasks and skills from our own lists; the server validates every pick, the user confirms it, and keyword matching replaces the model if it is unavailable; (2) "Refresh data", which fetches and validates METR's public data file, or keeps the last good copy and tells the user if that fails. Custom profiles stay in the browser's local storage.

**Hosting:** Replit, with the API key in Replit Secrets. A block diagram is in Appendix A.

### 2.2 Languages, Frameworks & Libraries

*List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.*

| Name & Version | Purpose in Project | License |
|---|---|---|
| JavaScript (ES modules), Node.js 22.22 | Language and runtime for API, engine and tests | MIT (Node.js) |
| Express 5.3.0 | REST API server | MIT |
| csv-parse 7.0.3 | Reads the CSV data snapshots | MIT |
| React and React DOM 19.3.1 | User interface | MIT |

| Name & Version | Purpose in Project | License |
|---|---|---|
| React Router DOM 7.18.4 | Page routing (You, Forecast, Map, Route, About) | MIT |
| Vite 8.3.4, @vitejs/plugin-react 6.1.2 | Dev server and production build | MIT |
| concurrently (^10.0.6), oxlint (^1.88.0) | Dev tools: run API and web together; linting | MIT |
| Node.js built-in test runner (node:test) | 111 automated engine tests | MIT (Node.js) |
| dotenv 18.0.8 | Loads settings (such as API keys) from environment files | BSD-2-Clause |
| js-yaml 5.5.0 | Used for METR | MIT |
| Plus Jakarta Sans | Typefaces | SIL OFL 1.1 |
| Bricolage Grotesque, DM Mono | Typefaces | SIL Open Font License 1.1 |

## 3. APIs & Cloud Services

### 3.1 External APIs

*List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.*

| API & Provider | Purpose in Project | Access Tier (free / paid / sponsor-provided) |
|---|---|---|
| Runpod Public Endpoint, Qwen3-32B-AWQ (Runpod) | Offline: one-off drafting of job-to-SDG links (2026-10-21). At runtime: optional free-text intake, which matches a worker's own words to tasks and skills from our lists. Output is validated by code and confirmed by the user. Text not saved. | Paid per use (voucher credits) |
| ESCO Web Service API (European Commission) | Offline: occupations, skills and job-skill links | Free |
| UN SDG API v5 (UN Statistics Division) | Offline: official text of the 169 SDG targets | Free |
| Microsoft Learn Catalog API | Offline: free courses and their stated durations | Free |
| Coursera public course catalogue API | Offline: courses and their stated workload | Free |
| METR benchmark results file (METR, public data over HTTPS) | Runtime: the Refresh button re-checks METR's published AI time-horizon data. It's validated, and if it fails we fall back to the last saved snapshot. | Free |

### 3.2 Cloud Platforms & Deployment

*List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.*

| Service & Provider | Purpose in Project | Plan / Region |
|---|---|---|
| Runpod (GPU inference) | Runs Qwen3-32B-AWQ: offline SDG-link drafting and the optional free-text intake at runtime | Pay-per-token public endpoint (hackathon credit, check) and the optional free-text intake at runtime |
| GitHub | Source code and data snapshots (tsetsen-oyun/SDC-Hackaton-2026) | Free plan, public repository. Region: not applicable |
| Replit | Hosting the web app and API | Autoscale deployment (2 vCPU / 4 GiB RAM), public; Replit Core |

# 4. AI Tools, Models & Datasets

## 4.1 AI Models Used in the Solution

*List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.*

| Model & Version | Source / Provider | Usage (pre-trained / fine-tuned / self-trained) | License / Terms |
|---|---|---|---|
| Qwen3-32B-AWQ | Qwen team (Alibaba), served through a Runpod public endpoint | Pre-trained, used as is. Offline: drafting SDG links (output stored as "Estimate"). At runtime: only the optional free-text intake (picks validated, user-confirmed). | Apache 2.0 |

The forecast does not use a machine-learning model of ours. It applies published METR time-horizon numbers and fixed rules, so any date can be traced to its source.

## 4.2 Datasets

*List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.*

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| O*NET Database 31.0 | onetcenter.org/database.html | public | CC BY 4.0 | N |
| Anthropic Economic Index (releases 2026-01-15, 2026-03-24, 2026-06-26) | huggingface.co/datasets/Anthropic/EconomicIndex | public | CC BY | N |

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| METR Time Horizon 1.1 and domains study | metr.org/time-horizons ; metr.org/blog/2025-07-14-how-does-time-horizon-vary-across-domains | public (numbers cited) | No license stated | N |
| Whitfill, Snodin and Becker, slowdown scenario | arxiv.org/abs/2511.19492 | public | CC BY 4.0; O*NET / European Commission | N |
| ESCO v1.0.9 (labels from v1.2.1) and ESCO to O*NET-SOC crosswalk | esco.ec.europa.eu; onetcenter.org/crosswalks.html | public | CC BY 4.0; O*NET / European Commission | N |
| Aurora SDG queries v5.0.3 | github.com/Aurora-Network-Global/sdg-queries | public | CC BY 4.0 | N |
| UN SDG targets (169) | unstats.un.org/sdgs/UNSDGAPIV5 | public | Free with citation | N |
| SDSN Sustainable Development Report 2026, Malaysia | dashboards.sdgindex.org/downloads | public | Permission by email from SDSN, 21 Oct 2026; non-commercial, cited | N |
| DOSM Job Market Insights Q2 2026 | dosm.gov.my (Big Data Analytics) | public | DOSM terms of use | N |
| US BLS Employment Projections 2025-35, Table 1.2 (US proxy) | bls.gov/emp/tables/occupational-projections-and-characteristics.htm | public | Public domain | N |

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| ISIS Malaysia and World Bank brief (45% figure) | Novel AI Technologies and the Future of Work in Malaysia, July 2025 | public (figure cited) | Published policy brief; one figure cited with attribution, not reproduced | N |
| Microsoft Learn and Coursera catalogues (96 courses) | learn.microsoft.com/api/catalog; api.coursera.org | public | Public catalogues; we link to each course | N |
| Team-made links: task-to-skill links, human review of SDG links | Team work (repository, data/docs) | self-collected | Team work (no AI) | N |
| Sample people (Aisyah, Daniel) | Built by the team from O*NET tasks | synthetic | Team work | N (not real people) |

### 4.3 AI-Assisted Development Tools

*Did your team use any generative-AI tool to help create code, design, text, or other project material? If yes, list them below:*

✓ Yes ☐ No

| **Tool & Version** | **Used For** | **Extent of Use (minor / moderate / major)** |
|---|---|---|
| Claude (Anthropic), Claude Sonnet 5.5 | Interface design and accessibility review, debugging. Animation of the video, no image generator was used. Only code. | Moderate for UI work; Minor for accessibility review; Moderate for debugging; Major for video generation. |
| Qwen3-32B-AWQ | Drafting job-to-SDG links (data, not code); see 4.1 | Major for the 516 model-drafted links, all labelled “Estimate” |
| ChatGPT (OpenAI) | Brainstorming, discussion assistance | Minor |

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

***[Maximum 100 words]***

Team members read, ran and edited all generated code before it entered the repository. The engine has 111 automated tests. We tested every page in a real browser against the real engine, including on a phone-sized screen. Model-drafted SDG links are labelled "Estimate" and "not yet checked" in the app until a team member reviews them; the team replaced rejected links by hand. Numbers shown to users come from cited sources, not from a model. The team leader is responsible for the final submission.

## 5. Sponsor-Provided Resources

*Did your project use any official SDC sponsor-provided resource?If yes, list them below:*

✓ Yes ☐ No

| **Sponsor** | **Resource (API / credits / platform / data)** | **How It Was Used in the Project** |
|---|---|---|
| RunPod | Public Endpoint API (Qwen3-32B-AWQ) and credits | Offline, once (2026-10-21): drafted job-to-SDG links, stored as labelled estimates. At runtime: optional free-text intake, where the model suggests tasks and skills from our own lists, every pick is checked by code, and keyword matching takes over if it is unavailable. It never produces forecasts, scores or rankings. |
| Replit | Cloud platform: Deployments (Autoscale, 2 vCPU / 4 GiB RAM) and Secrets | Hosts the live app (https://unbundle.replit.app/): one Express server serves the built React app and the API. The Runpod API key is kept in Replit Secrets, never in the repository. |

*Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:*

***[Maximum 100 words]***

Neither is essential. Forecast, matching, routing and mentoring are rule-based code that runs without any AI model. Runpod's Qwen3-32B drafted the SDG links once (already saved in our data) and powers the optional free-text intake; if it is unavailable, keyword matching takes over. Replit hosts the live app, but the app is a standard Node/Express server that could run on any host with the same build and start commands. Both saved us time but can be replaced.

## 6. Security, Privacy & Responsible Use

### 6.1 Data Handling & Privacy

*What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.*

***[Maximum 150 words]***

No user accounts, no sign-in and no real personal data. When a user changes hours or builds a custom profile, the API receives only task IDs, hours and skill IDs, calculates a result, and does not store or log them. Custom profiles stay in the user's own browser (local storage) and can be deleted from the You page. If a user types a description of their week into the optional intake box, that text is sent to the Runpod-hosted model to pick tasks and skills; it is not saved or logged, is limited to 3,000 characters, and users are asked to use made-up details. The browser loads fonts from Google Fonts. Refresh fetches METR's public file and sends no user data. The mentor match uses sample profiles only. CV text is read in the browser and never uploaded.

### 6.2 Known Limitations & Risks

*Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.*

***[Maximum 150 words]***

Forecast dates are estimates, not predictions. They scale METR's software-task time horizons by an assumed factor of 50 for office work, and Anthropic's usage data is not specific to Malaysia. Tasks with no published AI timing are counted as human work, which may understate exposure. Job outlook uses US BLS openings as a proxy. The 516 SDG links drafted by Qwen3-32B-AWQ may contain errors or bias; they are labelled "Estimate" and mostly unreviewed. HRD Corp claim status is not checked. The data covers 12 jobs and 96 courses. The optional free-text intake depends on one provider (Runpod), can be slow when idle, and could mis-pick; every pick is validated against our lists and confirmed by the user, requests are rate-limited, and keyword matching replaces the model if it fails. Each limit is shown on the About page.

## 7. Compliance & Readiness Checklist

*Tick all boxes that apply before submitting.*

- ✓ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ✓ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ✓ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ✓ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ✓ Any use of official SDC sponsor resources has been declared in Section 5.
- ✓ The prototype described here is accessible to judges through the links provided in Section 1.
- ✓ All fields in this template have been completed within the word limits stated.

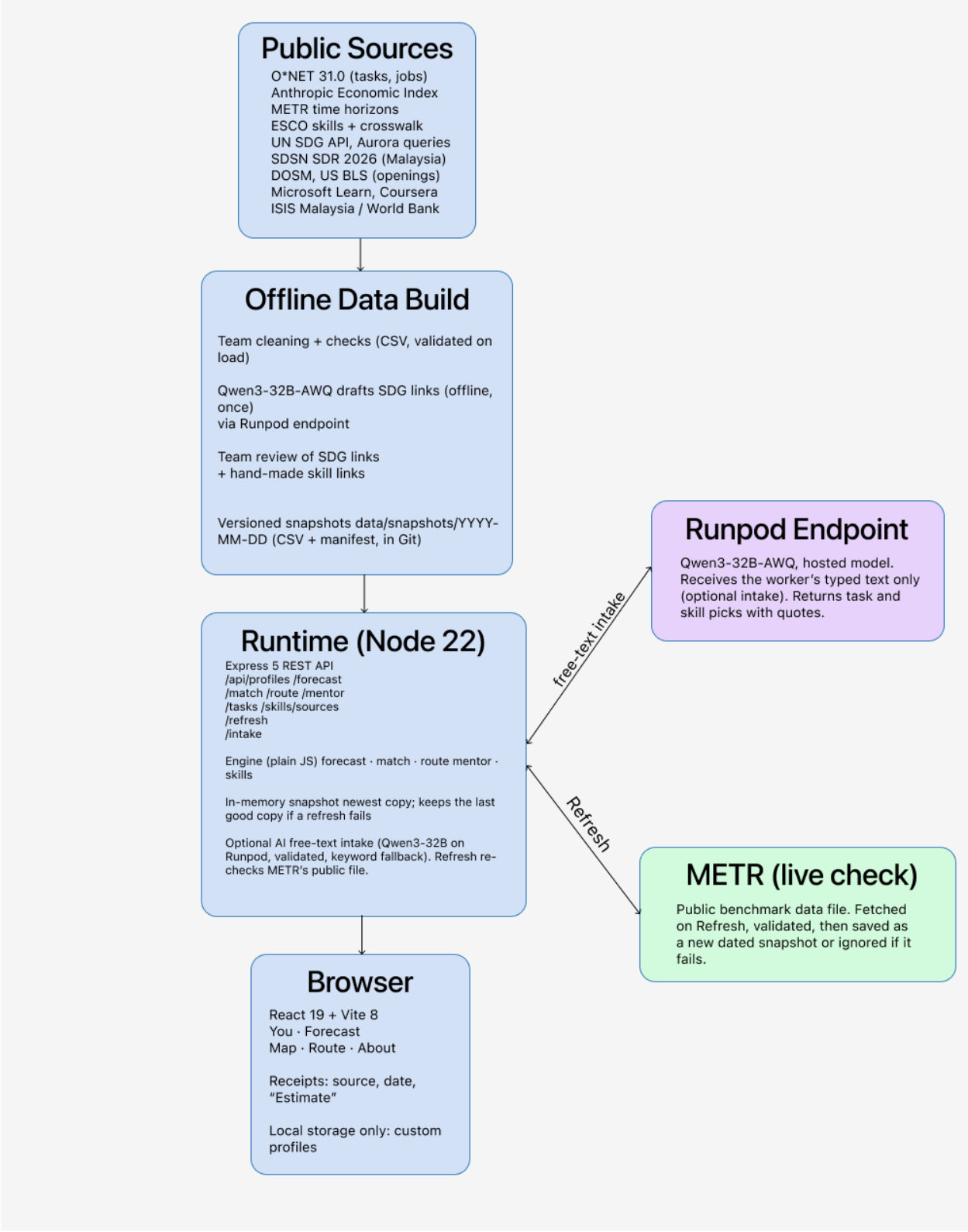
## 8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



25/10/2026

## Appendix A: System Block Diagram



Requests carry task IDs, hours and skill IDs; the optional intake also sends the typed text to the model. Nothing is stored on the server.

Left to right: public sources are cleaned offline into dated CSV snapshots. At runtime the Express API loads the newest snapshot into memory, the engine calculates results, and the React app shows them with a source, date and “Estimate” label on every number.